Multiple Linear Aggression

Q.1) Implement a linear regression algorithm to predict a continuous target variable based on more than one predictor variable. E.g. Implement a linear regression model on a housing prices dataset to predict the prices of houses based on their features

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample housing dataset
data = {
   'Area': [1000, 1200, 1500, 1800, 2000, 2200, 2500, 2700, 3000, 3500],
   'Bedrooms': [2, 2, 3, 3, 3, 4, 4, 4, 5, 5],
   'Bathrooms': [1, 2, 2, 2, 3, 3, 3, 4, 4, 4],
   'Price': [200000, 250000, 300000, 350000, 400000,
          450000, 500000, 550000, 600000, 700000]
}

# Create DataFrame
df = pd.DataFrame(data)

# Independent variables (features)
X = df[['Area', 'Bedrooms', 'Bathrooms']]

# Dependent variable (target)
y = df['Price']

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=42
)
```

```
# Create the Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict house prices
y_pred = model.predict(X_test)

# Display actual and predicted values
print("Actual Prices:", y_test.values)
print("Predicted Prices:", y_pred)

# Model coefficients
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)

# Predict price for a new house
new_house = pd.DataFrame({
   'Area': [2100],
   'Bedrooms': [3],
   'Bathrooms': [3]
})

predicted_price = model.predict(new_house)

print("Predicted Price of New House:", predicted_price[0])
```

Output of the code:-

```
Actual Prices: [600000 250000]
Predicted Prices: [610344.82758621 240086.20689655]
Intercept: -14224.137931036123
Coefficients: [  176.72413793  9913.79310345 11206.89655172]
Mean Squared Error: 102649375.74317661
R2 Score: 0.9966481836492024
Predicted Price of New House: 420258.62068965525
```